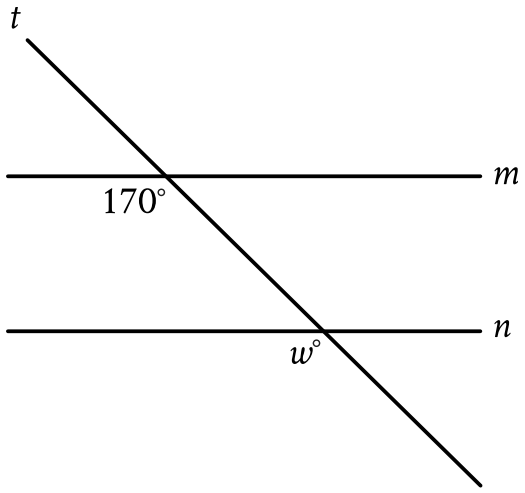
Question ID: 5207e508

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n. What is the value of w?

Answer

A. 17

B. 30

C. 70

D. 170

Correct Answer: D

Rationale

Choice D is correct. It's given that lines m and n are parallel. Since line t intersects both lines m and n, it's a transversal. The angles in the figure marked as $170°$ and $w°$ are on the same side of the transversal, where one is an interior angle with line m as a side, and the other is an exterior angle with line n as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that $w° = 170°$. Therefore, the value of w is 170.

Choice A is incorrect and may result from conceptual or calculation errors.

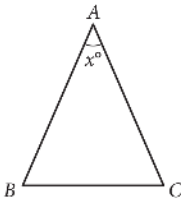
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: c8d60e48

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



In the given triangle, $AB = AC$ and $\angle ABC$ has a measure of $67°$. What is the value of x ?

Answer

A. 36

B. 46

C. 58

D. 70

Correct Answer: B

Rationale

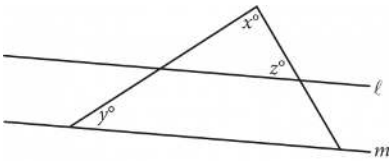
Choice B is correct. Since $AB = AC$, the measures of their corresponding angles, $\angle ABC$ and $\angle ACB$, are equal. Since $\angle ABC$ has a measure of $67°$, the measure of $\angle ACB$ is also $67°$. Since the sum of the measures of the interior angles in a triangle is $180°$, it follows that $67 + 67 + x = 180$, or $134 + x = 180$. Subtracting by 134 on both sides of this equation yields $x = 46$.

Choices A, C, and D are incorrect and may result from calculation errors.

Question ID: a6dbad6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure above, lines ℓ and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

Answer

A. 120

B. 100

C. 90

D. 80

Correct Answer: B

Rationale

Choice B is correct. Let the measure of the third angle in the smaller triangle be $a°$. Since lines ℓ and m are parallel and cut by transversals, it follows that the corresponding angles formed are congruent. So $a° = y° = 20°$. The sum of the measures of the interior angles of a triangle is $180°$, which for the interior angles in the smaller triangle yields $a + x + z = 180$. Given that $z = 60$ and $a = 20$, it follows that $20 + x + 60 = 180$. Solving for x gives $x = 180 - 60 - 20$, or $x = 100$.

Choice A is incorrect and may result from incorrectly assuming that angles $x + z = 180$. Choice C is incorrect and may result from incorrectly assuming that the smaller triangle is a right triangle, with x as the right angle. Choice D is incorrect and may result from a misunderstanding of the exterior angle theorem and incorrectly assuming that $x = y + z$.

Question ID: cbe8ca31

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question

In $\triangle XYZ$, the measure of $\angle X$ is $24°$ and the measure of $\angle Y$ is $98°$. What is the measure of $\angle Z$?

Answer

A. $58°$

B. $74°$

C. $122°$

D. $212°$

Correct Answer: A

Rationale

Choice A is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is $180°$. It's given that in $\triangle XYZ$, the measure of $\angle X$ is $24°$ and the measure of $\angle Y$ is $98°$. It follows that the measure of $\angle Z$ is $(180 - 24 - 98)°$, or $58°$.

Choice B is incorrect and may result from conceptual or calculation errors.

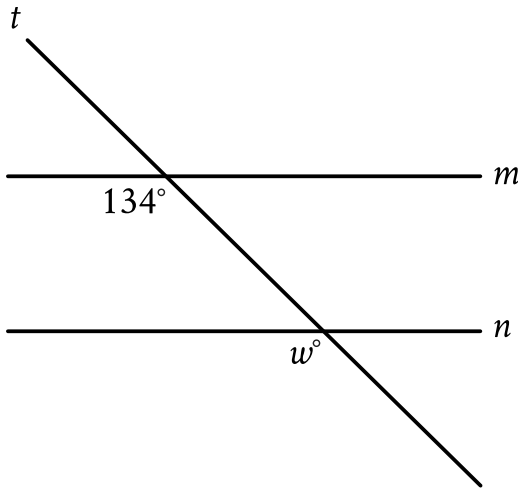
Choice C is incorrect. This is the sum of the measures of $\angle X$ and $\angle Y$, not the measure of $\angle Z$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: c24e1bda

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n. What is the value of w?

Answer

A. 13

B. 34

C. 66

D. 134

Correct Answer: D

Rationale

Choice D is correct. It's given that lines m and n are parallel. Since line t intersects both lines m and n, it's a transversal. The angles in the figure marked as $134°$ and $w°$ are on the same side of the transversal, where one is an interior angle with line m as a side, and the other is an exterior angle with line n as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that $w° = 134°$. Therefore, the value of w is 134.

Choice A is incorrect and may result from conceptual or calculation errors.

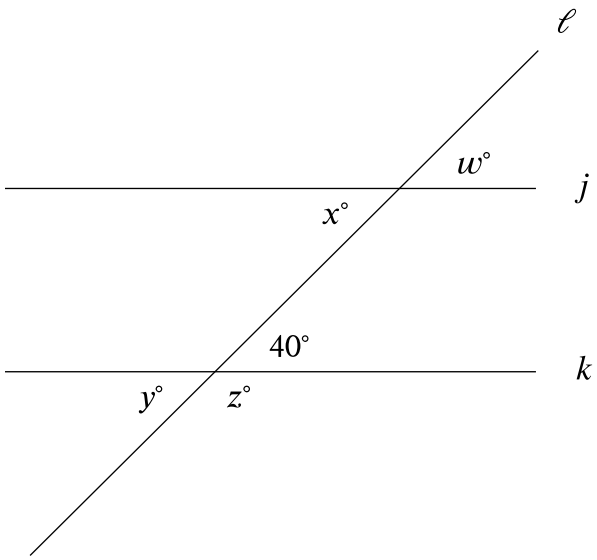
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 9d078710

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure shown, line ℓ intersects lines j and k. Which additional piece of information is sufficient to prove that lines j and k are parallel?

Answer

A. $w = 40$

B. $x = 140$

C. $y = 40$

D. $z = 140$

Correct Answer: A

Rationale

Choice A is correct. In the figure shown, lines j and k are parallel if and only if a pair of corresponding angles are congruent. It's given that one angle has a measure of $40°$ and that the corresponding angle has a measure of $w°$. Therefore, $w = 40$ is sufficient to prove that lines j and k are parallel.

Choice B is incorrect. The angle measuring $x°$ and the angle measuring $40°$ are alternate interior angles. Thus, if lines j and k are parallel, x is equal to 40, not 140.

Choice C is incorrect. The angle measuring $y°$ and the angle measuring $40°$ are vertical angles. Thus, $y = 40$, whether lines j and k are parallel or not.

Choice D is incorrect. The angle measuring $z°$ is supplementary to the angle measuring $40°$. Thus, $z = 180 - 40$, or $z = 140$, whether lines j and k are parallel or not.

Question ID: 3563d76d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question

At a certain time and day, the Washington Monument in Washington, DC, casts a shadow that is 300 feet long. At the same time, a nearby cherry tree casts a shadow that is 16 feet long. Given that the Washington Monument is approximately 555 feet tall, which of the following is closest to the height, in feet, of the cherry tree?

Answer

A. 10

B. 20

C. 30

D. 35

Rationale

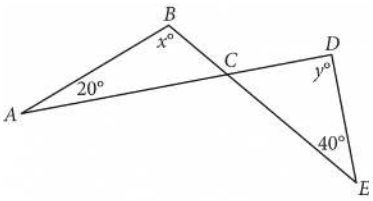
Choice C is correct. There is a proportional relationship between the height of an object and the length of its shadow. Let c represent the height, in feet, of the cherry tree. The given relationship can be expressed by the proportion $\frac{555}{300} = \frac{c}{16}$. Multiplying both sides of this equation by 16 yields $c = 29.6$. This height is closest to the value given in choice C, 30.

Choices A, B, and D are incorrect and may result from calculation errors.

Question ID: dfc420b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure above, $\overline{AD}$ intersects $\overline{BE}$ at C. If $x = 100$, what is the value of y ?

Answer

A. 100

B. 90

C. 80

D. 60

Correct Answer: C

Rationale

Choice C is correct. It's given that $x = 100$; therefore, substituting 100 for x in triangle ABC gives two known angle measures for this triangle.

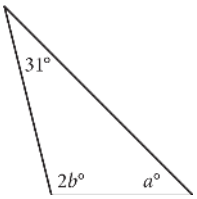
The sum of the measures of the interior angles of any triangle equals 180°. Subtracting the two known angle measures of triangle ABC from 180° gives the third angle measure: $180° - 100° - 20° = 60°$. This is the measure of angle BCA. Since vertical angles are congruent, the measure of angle DCE is also 60°. Subtracting the two known angle measures of triangle CDE from 180° gives the third angle measure: $180° - 60° - 40° = 80°$. Therefore, the value of y is 80.

Choice A is incorrect and may result from a calculation error. Choice B is incorrect and may result from classifying angle CDE as a right angle. Choice D is incorrect and may result from finding the measure of angle BCA or DCE instead of the measure of angle CDE.

Question ID: 410bdbe6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



In the triangle above, $a = 45$. What is the value of b ?

Answer

A. 52

B. 59

C. 76

D. 104

Correct Answer: A

Rationale

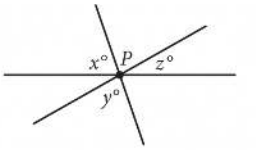
Choice A is correct. The sum of the measures of the three interior angles of a triangle is 180°. Therefore, $31 + 2b + a = 180$. Since it's given that $a = 45$, it follows that $31 + 2b + 45 = 180$, or $2b = 104$. Dividing both sides of this equation by 2 yields $b = 52$.

Choice B is incorrect and may result from a calculation error. Choice C is incorrect. This is the value of $a + 31$. Choice D is incorrect. This is the value of $2b$.

Question ID: 087cdcfd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, three lines intersect at point P. If $x = 65$ and $y = 75$, what is the value of z ?

Answer

A. 140

B. 80

C. 40

D. 20

Correct Answer: C

Rationale

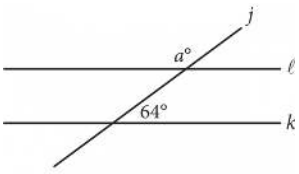
Choice C is correct. The angle that is shown as lying between the y° angle and the z° angle is a vertical angle with the x° angle. Since vertical angles are congruent and $x = 65$, the angle between the y° angle and the z° angle measures 65°. Since the 65° angle, the y° angle, and the z° angle are adjacent and form a straight angle, it follows that the sum of the measures of these three angles is 180°, which is represented by the equation $65° + y° + z° = 180°$. It's given that y = 75. Substituting 75 for y yields $65° + 75° + z° = 180°$, which can be rewritten as $140° + z° = 180°$. Subtracting 140° from both sides of this equation yields $z° = 40°$. Therefore, $z = 40$.

Choice A is incorrect and may result from finding the value of $x + y$ rather than z. Choices B and D are incorrect and may result from conceptual or computational errors.

Question ID: 992f4e93

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure above, lines ℓ and k are parallel. What is the value of a ?

Answer

A. 26

B. 64

C. 116

D. 154

Correct Answer: C

Rationale

Choice C is correct. Since lines ℓ and k are parallel, corresponding angles formed by the intersection of line j with lines ℓ and k are congruent. Therefore, the angle with measure a° must be the supplement of the angle with measure 64°. The sum of two supplementary angles is 180°, so a = 180 – 64 = 116.

Choice A is incorrect and likely results from thinking the angle with measure a° is the complement of the angle with measure 64°. Choice B is incorrect and likely results from thinking the angle with measure a° is congruent to the angle with measure 64°. Choice D is incorrect and likely results from a conceptual or computational error.

Question ID: f1747a6a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question

In triangle ABC, the measure of angle B is $52°$ and the measure of angle C is $17°$. What is the measure of angle A?

Answer

A. $21°$

B. $35°$

C. $69°$

D. $111°$

Correct Answer: D

Rationale

Choice D is correct. The sum of the angle measures of a triangle is $180°$. Adding the measures of angles B and C gives $52 + 17 = 69°$. Therefore, the measure of angle A is $180 - 69 = 111°$.

Choice A is incorrect and may result from subtracting the sum of the measures of angles B and C from $90°$, instead of from $180°$.

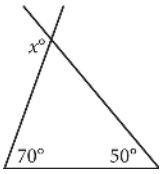
Choice B is incorrect and may result from subtracting the measure of angle C from the measure of angle B.

Choice C is incorrect and may result from adding the measures of angles B and C but not subtracting the result from $180°$.

Question ID: 36200a38

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



In the figure above, two sides of a triangle are extended. What is the value of x ?

Answer

A. 110

B. 120

C. 130

D. 140

Correct Answer: B

Rationale

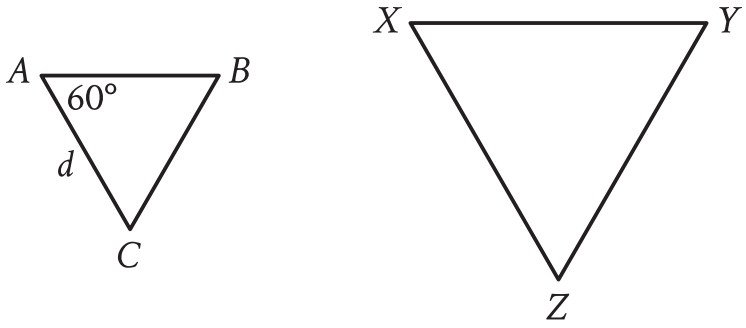
Choice B is correct. The sum of the interior angles of a triangle is 180°. The measures of the two interior angles of the given triangle are shown. Therefore, the measure of the third interior angle is 180° – 70° – 50° = 60°. The angles of measures x° and 60° are supplementary, so their sum is 180°. Therefore, x = 180 – 60 = 120.

Choice A is incorrect and may be the result of misinterpreting x° as supplementary to 70°. Choice C is incorrect and may be the result of misinterpreting x° as supplementary to 50°. Choice D is incorrect and may be the result of a calculation error.

Question ID: e0d2e21a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figures not drawn to scale.

For the triangles shown, triangle ABC is dilated by a scale factor of 3 to obtain triangle XYZ, where $d = 16$. What is the measure, in degrees, of angle X?

Answer

A. 20

B. 57

C. 60

D. 63

Correct Answer: C

Rationale

Choice C is correct. It's given that triangle XYZ is obtained by a dilation of triangle ABC. It follows that triangle ABC is similar to triangle XYZ, where A corresponds to X. Since corresponding angles in similar triangles have the same measure and the measure of angle A is 60 degrees, it follows that the measure of angle X is also 60 degrees.

Choice A is incorrect and may result from conceptual errors.

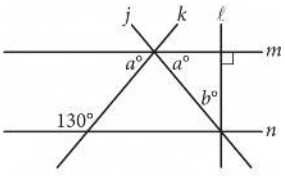
Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: 3828f53d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure above, lines m and n are parallel. What is the value of b ?

Answer

A. 40

B. 50

C. 65

D. 80

Correct Answer: A

Rationale

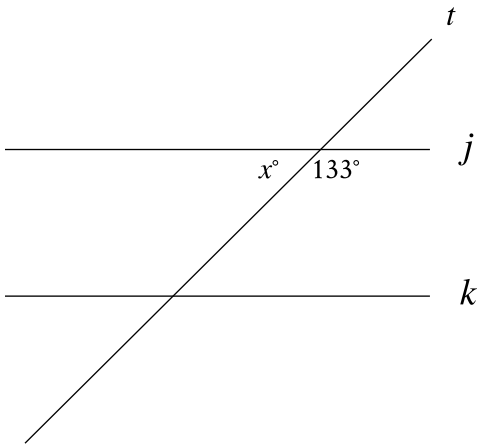
Choice A is correct. Given that lines m and n are parallel, the angle marked 130° must be supplementary to the leftmost angle marked a° because they are same-side interior angles. Therefore, 130° + a° = 180°, which yields a = 50°. Lines ℓ and m intersect at a right angle, so lines j, ℓ, and m form a right triangle where the two acute angles are a° and b°. The acute angles of a right triangle are complementary, so a° + b° = 90°, which yields 50° + b° = 90°, and b = 40.

Choice B is incorrect. This is the value of a, not b. Choice C is incorrect and may be the result of dividing 130° by 2. Choice D is incorrect and may be the result of multiplying b by 2.

Question ID: 3b4b5b1e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line j is parallel to line k. What is the value of x?

Correct Answer: 47

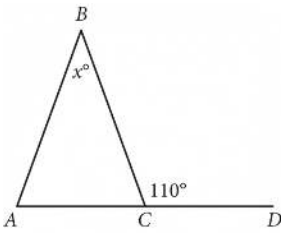
Rationale

The correct answer is 47. Based on the figure, the angle with measure $x°$ and the angle with measure $133°$ together form a straight line. Therefore, these two angles are supplementary, so the sum of their measures is $180°$. It follows that $x + 133 = 180$. Subtracting 133 from both sides of this equation yields $x = 47$.

Question ID: 5733ce30

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



In the given figure, $\overline{AC}$ extends to point D. If the measure of $\angle BAC$ is equal to the measure of $\angle BCA$, what is the value of x ?

Answer

A. 110

B. 70

C. 55

D. 40

Correct Answer: D

Rationale

Choice D is correct. Since $\angle BCD$ and $\angle BCA$ form a linear pair of angles, their measures sum to 180°. It's given that the measure of $\angle BCD$ is 110°. Therefore, $110° + \angle BCA = 180°$. Subtracting 110° from both sides of this equation gives the measure of $\angle BCA$ as 70°. It's also given that the measure of $\angle BAC$ is equal to the measure of $\angle BCA$. Thus, the measure of $\angle BAC$ is also 70°. The measures of the interior angles of a triangle sum to 180°. Thus, $70° + 70° + x° = 180°$. Combining like terms on the left-hand side of this equation yields $140° + x° = 180°$. Subtracting 140° from both sides of this equation yields $x° = 40°$, or $x = 40$.

Choice A is incorrect. This is the value of the measure of $\angle BCD$. Choice B is incorrect. This is the value of the measure of each of the other two interior angles, $\angle BCA$ and $\angle BAC$. Choice C is incorrect and may result from an error made when identifying the relationship between the exterior angle of a triangle and the interior angles of the triangle.